Low Loss **High Loss**

Depth [km]

a)

$d_{porous\ layer}$ = 250 m,
$k_{porous\ layer}$ = 0.025 W/(mK)
d_{ice} = 5 km
d_{ice} = 21 km
d_{ice} = 35 km

d)

Ice Shell Thickness
5 km
21 km
35 km

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

g)

Ice Shell Thickness
5 km
21 km
35 km

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

b)

d_{ice} = 21 km,
$k_{porous\ layer}$ = 0.025 W/(mK)
$d_{porous\ layer}$ = 0 m
$d_{porous\ layer}$ = 100 m
$d_{porous\ layer}$ = 400 m
$d_{porous\ layer}$ = 700 m

e)

Porous Layer Thickness
0 meters
100 meters
400 meters
700 meters

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

h)

Porous Layer Thickness
0 meters
100 meters
400 meters
700 meters

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

c)

d_{ice} = 21 km,
$d_{porous\ layer}$ = 250 m
$k_{porous\ layer}$ = 0.1 W/(mK)
$k_{porous\ layer}$ = 0.025 W/(mK)
$k_{porous\ layer}$ = 0.001 W/(mK)

f)

Porous Layer Conductivity
0.001 W/(mK)
0.025 W/(mK)
0.1 W/(mK)

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

i)

Porous Layer Conductivity
0.001 W/(mK)
0.025 W/(mK)
0.1 W/(mK)

2-way Attenuation
< 1 dB
40 dB
80 dB
> 100 dB

Depth [km]

Temperature [K]